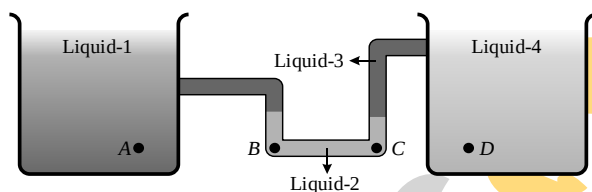


JEE Main Level Practice Test-10

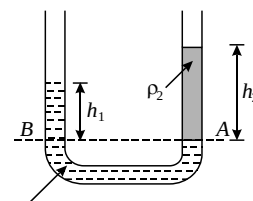
for JEE & NEET Aspirants

Topic : FLUID STATICS & DYNAMICS
Time: 75 Min Marking +4 -1
Section - A : MCQs with Single Option Correct

- A balloon of volume 20 m^3 is filled with helium. It rises to a height of 180 m in 30 s starting from rest. The mass of balloon and equipment (excluding payload and helium) is 12 kg. If density of air is 1.29 kg/m^3 and density of helium is 0.18 kg/m^3 and are constant, find the mass of payload lifted by the balloon :
 (A) 7.5 kg (B) 8 kg (C) 8.8 kg (D) 9.2 kg
- Which of the following is correct for the given diagram ? A, B, C and D are in same horizontal level.



- (A) Pressure at A = pressure at B
 (B) Pressure at B = pressure at C
 (C) Pressure at C = pressure at D
 (D) All of the above
- Two immiscible liquids are poured in a U-tube having densities ρ_1 and ρ_2 . The ratio of height of the liquids above their interface (h_1/h_2) is :
 (A) Directly proportional to their densities
 (B) Inversely proportional to their densities
 (C) Directly proportional to square of their densities
 (D) Equal



- A sphere of volume V is floating in water kept in a vessel such that half of its volume is submerged in water. Now an immiscible liquid whose density is lesser than water as well as sphere is poured over the water. The sphere is now completely submerged. The volume of sphere submerged in :
 (A) Water is more than $\frac{V}{2}$ but in liquid is less than $\frac{V}{2}$
 (B) Water is less than $\frac{V}{2}$ but in liquid is more than $\frac{V}{2}$
 (C) Water as well as liquid is $\frac{V}{2}$
 (D) The sphere will not have any part submerged in water
- A wooden block floating in a bucket of water has $4/5$ of its volume submerged. When certain amount of an oil is poured into the bucket, it is found that the block is just under the oil surface with half of its volume under water and half in oil. The density of oil relative to that of water is :
 (A) 0.5 (B) 0.7 (C) 0.6 (D) 0.8

6. A cubical block of side 0.5 m floats on water with 30% of its volume under water. What is the maximum weight that can be put on the block without fully submerging it under water ? (Take density of water = 10^3 kg/m^3)
 (A) 65.4 kg (B) 87.5 kg (C) 30.1 kg (D) 46.3 kg

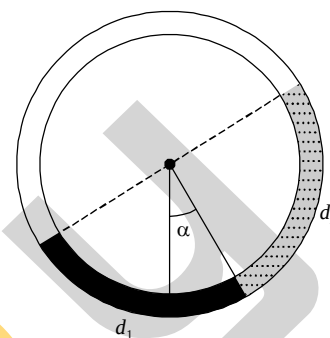
7. A uniform cylinder of length L and mass M having cross-sectional area A is suspended, with its length vertical, from a fixed point by a massless spring, such that it is half submerged in a liquid of density σ at equilibrium position. The extension x_0 of the spring when it is in equilibrium is :

(A) $\frac{Mg}{k} \left(1 - \frac{LA\sigma}{M} \right)$ (B) $\frac{Mg}{k} \left(1 - \frac{LA\sigma}{2M} \right)$ (C) $\frac{Mg}{k} \left(1 + \frac{LA\sigma}{M} \right)$ (D) $\frac{Mg}{k}$

8. There is a circular tube in a vertical plane. Two liquids which do not mix and of densities d_1 and d_2 are filled in the tube. Each liquid subtends 90° angle at centre.

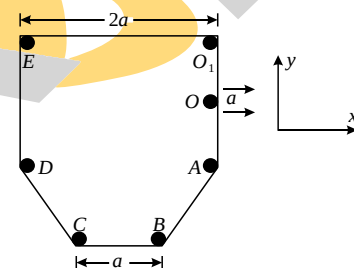
Radius joining their interface makes an angle α with vertical. Ratio $\frac{d_1}{d_2}$ is :

(A) $\frac{1 + \sin \alpha}{1 - \sin \alpha}$ (B) $\frac{1 + \cos \alpha}{1 - \cos \alpha}$
 (C) $\frac{1 + \tan \alpha}{1 - \tan \alpha}$ (D) $\frac{1 + \sin \alpha}{1 - \cos \alpha}$



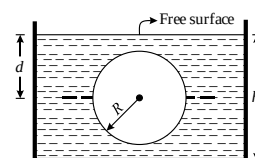
9. A closed compartment containing liquid (as shown) is moving with an acceleration along x -axis. Find the incorrect statement. (ignore gravity)

- (A) The pressure at A and O is same
 (B) The pressure at O and O_1 is same
 (C) The pressure at B and C is same
 (D) The pressure at D and E is same



10. A uniform solid sphere of radius R is in equilibrium inside a liquid whose density varies with depth from free surface as $\rho = \rho_0 \left(1 + \frac{h}{h_0} \right)$. Where h is depth from free surface. Density of sphere σ will be :

(A) $\sigma = \rho_0 \left(1 + \frac{d}{2h_0} \right)$ (B) $\sigma = \rho_0 \left(1 - \frac{d}{2h_0} \right)$
 (C) $\sigma = \rho_0 \left(1 + \frac{2d}{h_0} \right)$ (D) $\sigma = \rho_0 \left(1 + \frac{d}{h_0} \right)$



11. A cylindrical vessel of cross-section A contains water to a height h . There is a hole in the bottom of radius ' a '. The time in which it will be emptied is :

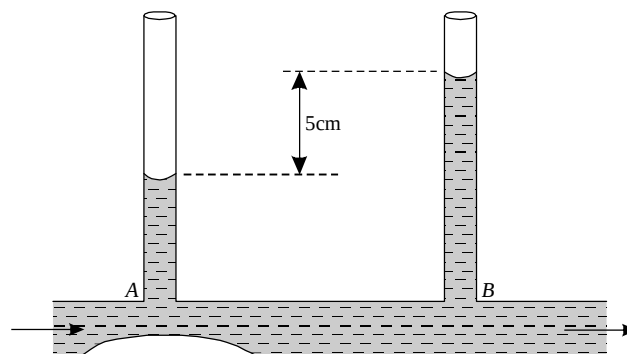
(A) $\frac{2A}{\pi a^2} \sqrt{\frac{h}{g}}$ (B) $\frac{\sqrt{2}A}{\pi a^2} \sqrt{\frac{h}{g}}$ (C) $\frac{2\sqrt{2}A}{\pi a^2} \sqrt{\frac{h}{g}}$ (D) $\frac{A}{\sqrt{2}\pi a^2} \sqrt{\frac{h}{g}}$

12. A tank with a small hole at the bottom has been filled with water and kerosene (specific gravity 0.8). The height of water is 3 m and that of kerosene 2 m. When the hole is opened the velocity of fluid coming out from it is nearly : (take $g = 10 \text{ ms}^{-2}$ and density of water = 10^3 kg m^{-3})

(A) 10.7 ms^{-1} (B) 9.6 ms^{-1} (C) 8.5 ms^{-1} (D) 7.6 ms^{-1}

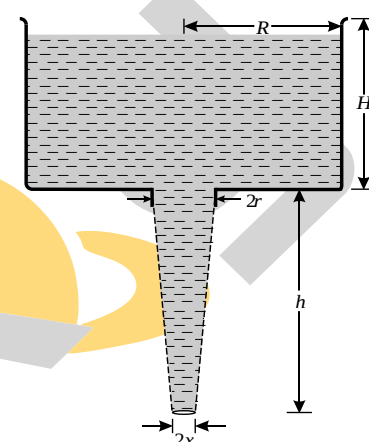
13. In the diagram shown, the difference in the two tubes of the manometer is 5 cm, the cross section of the tube at A and B is 6 mm^2 and 10 mm^2 respectively. The rate at which water flows through the tube is ($g = 10 \text{ ms}^{-2}$):

(A) 7.5 cc/s
(B) 8.0 cc/s
(C) 10.0 cc/s
(D) 12.5 cc/s



14. Consider a water jar of radius R that has water filled up to height H and is kept on a stand of height h (see figure). Through a hole of radius r ($r \ll R$) at its bottom, the water leaks out and the stream of water coming down towards the ground has a shape like a funnel as shown in the figure. If the radius of the cross-section of water stream when it hits the ground is x . Then :

(A) $x = r \left(\frac{H}{H+h} \right)^{\frac{1}{4}}$ (B) $x = r \left(\frac{H}{H+h} \right)$
(C) $x = r \left(\frac{H}{H+h} \right)^2$ (D) $x = r \left(\frac{H}{H+h} \right)^{\frac{1}{2}}$



15. The top of a water tank is open to air and its water level is maintained. It is giving out 0.74 m^3 water per minute through a circular opening of 2 cm radius in its wall. The depth of the centre of the opening from the level of water in the tank is close to :

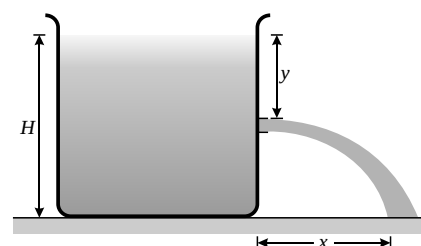
(A) 9.6m (B) 4.8m (C) 2.9m (D) 6.0m

16. A liquid of density ρ is coming out of a hose pipe of radius a with horizontal speed v and hits a mesh. 50% of the liquid passes through the mesh unaffected. 25% loses all of its momentum and 25% comes back with the same speed. The resultant pressure on the mesh will be :

(A) ρv^2 (B) $\frac{3}{4} \rho v^2$ (C) $\frac{1}{2} \rho v^2$ (D) $\frac{1}{4} \rho v^2$

17. A tank which is open at the top contains a liquid upto a height H . A small hole is made in the side of the tank at a distance y below the liquid surface. The liquid emerging from the hole lands at a horizontal distance x from the tank. Choose incorrect option :

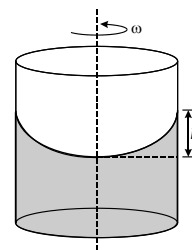
(A) If y is increased from zero to H , x will first increase and then decrease
(B) x is maximum for $y = H/2$
(C) The maximum value of x is H
(D) The maximum value of x will depend on the density of liquid



18. Water stands upto a height h behind the vertical wall of a dam. What is the net horizontal force pushing the dam down by the stream, if width of the dam is σ ? (ρ = density of water) :

(A) $2h\sigma g$ (B) $\frac{h^2\sigma\rho g}{2}$ (C) $\frac{h^2\sigma\rho g}{4}$ (D) $\frac{h\sigma\rho g}{4}$

19. A water bucket filled with water is rotating with angular velocity of 10 rad s^{-1} as shown. Radius of bucket is 1 m. What is the value of 'h'? (Take $g = 10 \text{ ms}^{-2}$)
- (A) 5m
(B) 1m
(C) 0.5m
(D) 0.25m



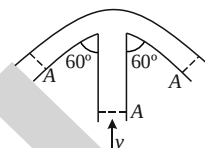
20. An ideal liquid of density ρ is pushed with velocity v through the central limb of the tube shown in the figure. What force does the liquid exert on the tube? The cross-sectional areas of the three limb are equal to A each. Assume streamline flow.

(A) $\frac{9}{8} \rho A v^2$

(B) $\frac{5}{4} \rho A v^2$

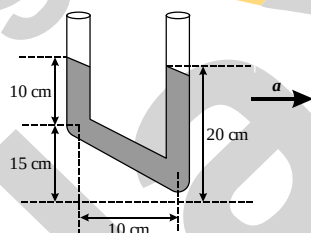
(C) $\frac{3}{2} \rho A v^2$

(D) $\rho A v^2$

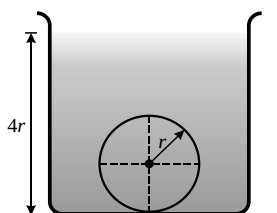


Section- B: INTEGER Answer Type Questions

21. A U tube with a tilted base arm filled with a liquid is accelerating horizontally with an acceleration a . The acceleration (m/s^2) of the tube is :

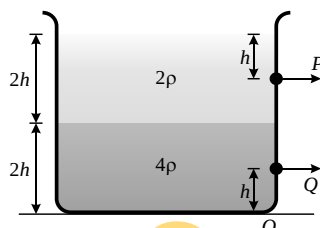


22. A block of wood has a mass of 25 g. When a 5 g metal piece with a volume of 2 cm^3 is attached to the bottom of the block, the wood barely floats in water. Calculate the volume (in cm^3) of the wooden block?
23. A submarine experiences a pressure of $5.05 \times 10^6 \text{ Pa}$ at a depth of d_1 in a sea. When it goes further to a depth of d_2 , it experiences a pressure of $8.08 \times 10^6 \text{ Pa}$. Then $d_2 - d_1$ (m) is approximately (density of water = 10^3 kg/m^3)
24. A small spherical metal ball of radius r is lying at the bottom of a stationary container containing liquid of density ρ as shown in the figure. P_0 is atmospheric pressure. Force exerted by liquid on the upper hemispherical part of the sphere is $\frac{\pi r^2}{3} [3P_0 + nr\rho g]$. Find n .

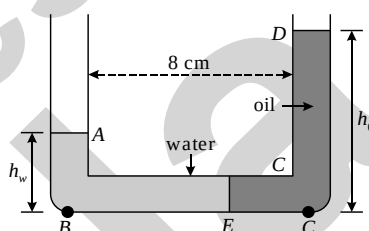


25. In a simple hydraulic press, the cross-sectional area of the two cylinders are $5 \times 10^{-4} \text{ m}^2$ and 10^{-2} m^2 respectively. A force of 20 N is applied at the small plunger. Calculate the pressure (kN/m^2) produced in the cylinders?

26. Water flows into a large tank with flat bottom at the rate of $10^{-4} \text{ m}^3 \text{ s}^{-1}$. Water is also leaking out of a hole of area 1 cm^2 at its bottom. If the height of the water in the tank remains steady, calculate this height (in cm).
27. A long cylindrical vessel is half filled with a liquid. When the vessel is rotated about its own vertical axis, the liquid rises up near the wall. If the radius of vessel is 5 cm and its rotational speed is 2 rotations per second, then calculate the difference in the heights between the centre and the sides in cm.
28. A large tank is filled with two liquids of specific gravities 2ρ and 4ρ . Two holes are made on the wall of the tank as shown. The ratio of the distance from O of the points on the ground where the jets from holes P & Q strike is $\sqrt{3} : \sqrt{n}$. Find n .



29. A tube of uniform cross-section has two vertical portions connected with a horizontal thin tube 8 cm long at their lower ends. Enough water to occupy 22 cm of the tube is poured into one branch and enough oil of specific gravity 0.8 to occupy 22 cm is poured into the other. Find the distance BE (in cm) of the common surface E of the two liquids.



30. The pressure of water in a water pipe when tap is open and closed are respectively $3 \times 10^5 \text{ Nm}^{-2}$ and $3.5 \times 10^5 \text{ Nm}^{-2}$. With open tap, calculate the velocity (in m/s) of water flowing from the tap. (density of water = 1000 kg/m^3)

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JEE Main Level Practice Test-10

for JEE & NEET Aspirants

Topic : FLUID STATICS & DYNAMICS
ANSWER KEY
Section - A : MCQs with Single Option Correct

- | | | | |
|---------|---------|---------|---------|
| 1. (D) | 2. (B) | 3. (B) | 4. (B) |
| 5. (C) | 6. (B) | 7. (B) | 8. (C) |
| 9. (C) | 10. (D) | 11. (B) | 12. (B) |
| 13. (A) | 14. (A) | 15. (B) | 16. (B) |
| 17. (D) | 18. (B) | 19. (A) | 20. (B) |

Section- B: INTEGER Answer Type Questions

- | | | | |
|----------|----------|-----------|---------|
| 21. [5] | 22. [28] | 23. [300] | 24. [7] |
| 25. [40] | 26. [5] | 27. [2] | 28. [2] |
| 29. [6] | 30. [10] | | |

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